

Determining Language Grounding Failures in Vision-Language-Action Models

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Abstract

Vision-Language-Action (VLA) models report high aggregate success rates on manipulation benchmarks, but these single-number metrics hide whether models genuinely follow instructions or exploit visual shortcuts and memorized scene layouts. In this paper, we isolate the language channel using a scene-fixed causal evaluation protocol on `LIBERO-Goal`, holding the initial simulator state strictly constant while perturbing instructions across three architectures spanning a $16\times$ parameter range (`SmolVLA-450M`, `OpenVLA-7B`, and `OpenVLA-OFT-7.5B`). We find that language is causally necessary and scale-invariant: destroying the instruction by blanking or scrambling it collapses task success across all scales. However, we show that widely reported word-class asymmetries—where models appear to bind object nouns strictly while treating action verbs as near-optional—are confounded by benchmark information structure. In `LIBERO-Goal`, the noun set uniquely identifies all ten tasks ($H(\text{task} \mid \text{nouns}) = 0$ bits), rendering the verb redundant. When we test genuine verb antonyms, `OpenVLA-7B` drops 31.7 points ($p < 10^{-5}$), proving that it reads the verb when the benchmark requires it. We present an episode-level goal predicate taxonomy to distinguish instruction-following from scene-memorized defaults, and argue that benchmark information content must be reported before declaring models instruction-blind.

1 Introduction

Vision-Language-Action (VLA) models combine vision-language backbones with robotic control heads, promising policies that execute physical manipulations driven by natural-language instructions (Brohan et al., 2022, 2023; Driess et al., 2023; Kim et al., 2025b; Shukor et al., 2025). Standard evaluation benchmarks report high aggregate

success rates, and the community frequently interprets these figures as evidence that policies genuinely understand their instructions. However, a model can score well on a benchmark without parsing language at all by exploiting visual shortcuts, memorized task layouts, or superficial keyword matching. This gap between aggregate success and genuine language grounding is the central problem we address in this paper.

We approach this problem through two core research questions:

- **RQ1:** Does language causally drive a VLA model’s physical behavior when the initial visual scene is held strictly constant?
- **RQ2:** Which linguistic operations break grounding, and do observed failures stem from object reference (nouns) or action reference (verbs)?

To answer these questions, we hold the physical simulator scene fixed and treat only the instruction text as the experimental variable. This design choice separates our work from existing VLA robustness benchmarks, which vary multiple factors simultaneously—such as object positions, camera angles, lighting, sensor noise, and language—or compound several perturbations at once. By isolating language as the sole manipulated variable, we determine the instruction’s exact role in driving policy behavior rather than making an aggregate robustness claim.

Evaluating three model architectures spanning a $16\times$ parameter range on `LIBERO-Goal`, we find that language is causally load-bearing and scale-invariant: destroying the instruction via blanking or length-matched nonsense collapses task success across all model scales. However, when we investigate word-class sensitivity, we discover that the widely reported asymmetry—where models seem sensitive to object nouns but insensitive to

action verbs—is an artifact of benchmark information structure. In **LIBERO-Goal**, the noun set carries all 3.32 bits of task identity ($H(\text{task} \mid \text{nouns}) = 0$ bits), rendering the verb fully redundant. When we isolate the three tasks with true verb antonyms, **OpenVLA-7B** loses 31.7 points ($p < 10^{-5}$), demonstrating that it reads the verb when the benchmark requires it. We propose an episode-level goal-predicate taxonomy and metric suite to separate genuine instruction compliance from scene-memorized defaults.

2 Preliminaries

2.1 Vision-Language-Action Models & Checkpoints

VLA models extend vision-language models with action heads to output motor controls conditioned on language instructions (Brohan et al., 2022, 2023; Driess et al., 2023; Collaboration et al., 2023). We evaluate three open-source checkpoints spanning a $16\times$ parameter range and three distinct action parameterizations: (1) **SmolVLA-450M** (Shukor et al., 2025), a compact policy combining SmolVLM2 with a flow-matching action expert; (2) **OpenVLA-7B** (Kim et al., 2025b), emitting discretized action tokens from a static camera; and (3) **OpenVLA-OFT-7.5B** (Kim et al., 2025a), utilizing continuous action chunking via L1 regression, an additional wrist camera, and 8D proprioception.

2.2 Benchmarks

We conduct all evaluations within the **LIBERO** benchmarking framework (Liu et al., 2023). **LIBERO** contains four distinct task suites: **Spatial**, **Object**, **Goal**, and **Long**. We focus exclusively on **LIBERO-Goal** because its ten tasks share an identical initial scene layout and object inventory (Figure 1a). In **LIBERO-Goal**, the initial visual state S_0 does not disambiguate the goal, making the instruction the sole distinguishing input.

3 Related Work

VLA Robustness and Benchmark Shortcuts.

Recent studies highlight severe vulnerabilities in VLA policies. Pang et al. (2025) evaluate the positional robustness of **OpenVLA** on **LIBERO-Object**, finding that success drops sharply outside the specific object positions seen during training, indicating rote memorization.

LIBERO-Plus (Fei et al., 2025) benchmarks over 10,000 tasks across seven variation axes, reporting that language perturbations cause minor performance drops while visual perturbations disrupt policies severely. **LIBERO-Para** (Kim et al., 2026) evaluates paraphrase robustness across object and action references and introduces the **PRIDE** metric, which we adopt to measure paraphrase sensitivity.

The Instruction-Blindness Consensus. A growing literature argues that VLAs largely ignore language. **LIBERO-PRO** (Zhou et al., 2026) reports that outputs remain unchanged under corrupted instructions, attributing success to memorized action sequences. **LangGap** (Hou and Zhao, 2026) demonstrates that high-scoring models fail when forced to use language. Zhang and Li (2026) term this failure mode “instruction blindness”, while Zhan and Wang (2026) report a “modality collapse” toward visual affordances. Mechanistically, Grant et al. (2026) show that injecting baseline activations into null-prompt episodes recovers baseline behavior across six models.

Linguistic Shortcuts in NLP. Reliance on superficial dataset shortcuts is a well-documented challenge in computational linguistics. In Natural Language Inference (NLI), models routinely exploit lexical overlap and syntactic heuristics (Gururangan et al., 2018; McCoy et al., 2019). In Visual Question Answering (VQA), models frequently exploit language priors to answer without inspecting image details (Goyal et al., 2017). Our work connects this NLP dataset-artifact literature to VLA evaluation, demonstrating that apparent “verb blindness” in VLAs is an artifact of information redundancy in benchmark instruction design.

4 Language Grounding Failures

4.1 Defining Language Grounding Failure

We define a *language grounding failure* as an episode in which the policy’s physical behavior is not causally driven by the linguistic content of the instruction. We distinguish this from an *execution failure*, in which the policy correctly infers the intended goal but fails to physically execute the trajectory.

Scalar success rates conflate these two independent properties: whether the policy understood the instruction, and whether the policy possessed the

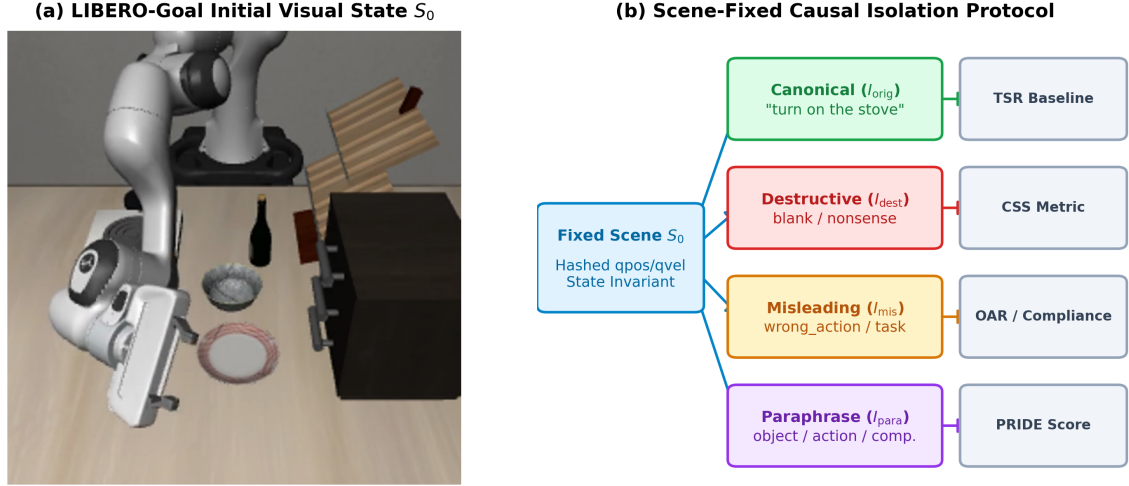


Figure 1: **LIBERO-Goal Environment Setup and Scene-Fixed Causal Isolation Framework.** (a) Top-down visual camera view of initial simulator state S_0 , depicting fixed tabletop object positions (cabinet, middle drawers, stove knob, bowl, plate, Panda gripper) shared identically across all ten tasks. (b) Conceptual diagram of our scene-fixed causal evaluation protocol: holding state S_0 strictly constant while perturbing instruction text to evaluate final simulator goal predicates $\mathcal{A}(S_T)$.

motor skill to complete it. A model can fail a task despite perfect language grounding (producing a correct approach trajectory that slips at the grasp phase). Conversely, a model can succeed at a task despite zero language grounding if its behavior is driven by memorized scene layouts or visual priors.

We resolve this ambiguity by holding the initial simulator state S_0 strictly constant while perturbing the instruction I . We then evaluate the final simulator state against all goal predicates achievable in that scene, rather than relying solely on the single predicate associated with the original task.

4.2 The Perturbation Taxonomy

Let I_{orig} denote the canonical instruction for a task on fixed scene S_0 . We construct a family of perturbed instructions I_{cond} grouped by the shortcut hypothesis each condition falsifies:

- **Destructive Conditions:** Remove instruction content entirely while leaving scene S_0 untouched. `blank` supplies an empty string; `nonsense` supplies length-matched out-of-domain tokens. If the policy relies on language, success must collapse. If the policy succeeds, its behavior is driven by scene memorization.
- **Misleading Conditions:** Substitute a valid, executable instruction describing a different

goal in S_0 . `wrong_object` replaces the target object noun; `wrong_action` replaces the action verb; `wrong_task` replaces the full instruction with another task’s instruction. These conditions test whether a policy follows the new instruction (compliance) or defaults to the scene’s original goal (memorization).

- **Control & Paraphrase Conditions:** `repeated` concatenates I_{orig} with itself to test sensitivity to length. Paraphrase conditions (`para_object`, `para_action`, `para_compositional`) reword the instruction along specific linguistic axes while preserving meaning, testing robustness to surface form.

Table 1 summarizes the perturbation taxonomy.

4.3 Ground-Truth Evaluation via Goal Predicates

Standard simulators evaluate rollouts using a single boolean flag corresponding to the original task predicate g_{k*} . Under perturbed instructions, this creates a mismatch: a policy that correctly follows a misleading instruction (e.g., executing `wrong_task`) is mis-scored as a failure by the native simulator check.

We exploit a structural property of LIBERO-Goal: all ten tasks are defined

Condition	Construction	Shortcut Hypothesis Ruled Out
original	I_{orig} , unchanged	Baseline reference.
blank	Empty string	Policy succeeds without language (pure scene memorization).
nonsense	Length-matched out-of-domain tokens	Policy relies on keyword presence without structural semantics.
wrong_object	Swapped target object noun	Policy ignores named object and defaults to a fixed target.
wrong_action	Swapped action verb	Policy ignores named action and defaults to a fixed manipulation.
wrong_task	Full instruction of another in-scene task	Policy ignores instruction and executes scene’s default task.
repeated	I_{orig} concatenated with itself	Sensitivity to input length rather than meaning.
para_object, para_action, para_compositional	Reworded instruction preserving original goal	Policy is brittle to surface form despite unchanged semantics.

Table 1: **Instruction Perturbation Taxonomy.** Classification of experimental conditions applied to canonical task instructions I_{orig} on fixed initial scene S_0 . Each perturbation type is designed to isolate and rule out a specific candidate shortcut strategy or ungrounded failure mode in Vision-Language-Action policies.

over the identical initial scene S_0 . We evaluate every task’s goal predicate g_k against the final state S_T of every rollout, constructing the *achieved task set*:

$$\mathcal{A}(S_T) = \{k : g_k(S_T) = \text{true}\} \quad (1)$$

where $k \in \{1, \dots, 10\}$. $\mathcal{A}(S_T)$ reveals whether the policy achieved no goal ($\emptyset$), the original goal (k^*), or a newly instructed goal (k_{wrong}).

4.4 Evaluating Perturbed Instructions via Achieved Goal Sets

Using $\mathcal{A}(S_T)$, we classify episode outcomes under perturbed prompts:

- **Destructive Conditions (blank, nonsense):** An episode is *Grounded* if I_{orig} succeeds but $\mathcal{A}(S_T) = \emptyset$ (removing language eliminates success). It is categorized as *Memorization* if $\mathcal{A}(S_T) \neq \emptyset$ despite missing language.
- **Misleading Conditions (wrong_*):** When I_{wrong} specifies goal $k_{\text{wrong}} \neq k^*$, we measure whether the policy complies with the new goal (*Follow*, $k_{\text{wrong}} \in \mathcal{A}(S_T)$), defaults to the scene’s original goal (*Ignore*, $k^* \in \mathcal{A}(S_T)$), or fails both (*Fail*).

4.5 Metrics

We convert episode classifications into four aggregate metrics: (1) **Causal Sensitivity Score (CSS):** For destructive conditions, $\text{CSS}(c) = \frac{\text{TSR}_0 - \text{TSR}(c)}{\max(\text{TSR}_0, \epsilon)}$, where $\text{CSS} \approx 1$ indicates strong causal necessity;

(2) **Original-Action Rate (OAR):** For misleading conditions, $\text{OAR}(c) = \frac{|\text{Ignore}|}{|\text{Follow}| + |\text{Ignore}| + |\text{Fail}|}$, measuring scene-memorization overrides; (3) **PRIDE Score:** For paraphrases, task success weighted by paraphrase distance (Kim et al., 2026); and (4) **Failure Locus:** Categorizing failures into *planning* vs. *execution* breakdowns.

5 Experiments

5.1 Models and Benchmark

We evaluate SmolVLA-450M, OpenVLA-7B, and OpenVLA-OFT-7.5B on the ten tasks of LIBERO-Goal in MuJoCo. All tasks share an identical initial scene layout and object inventory (Figure 1a), ensuring that the instruction is the sole distinguishing input.

5.2 Scene-Fixed Protocol

For each (task, episode) pair, we hold the initial simulator state S_0 strictly constant across all instruction conditions: $S_0(I_{\text{orig}}) \equiv S_0(I_{\text{pert}})$. We audit this invariant by hashing simulator state arrays (`qpos`, `qvel`) immediately after environment reset. As detailed in Section B, un-audited evaluations frequently violate this invariant due to simulator generator drift across process resets.

5.3 Data Collection and Implementation

Each rollout logs a structured record containing model checkpoint hash, task ID, condition, initial-state hash, achieved task set $\mathcal{A}(S_T)$, and end-effector kinematics. Rollouts run for $T = 300$ max steps across independent randomized evaluations. Simulator evaluations execute sequentially

with one EGL rendering context per process to prevent rendering context crashes.

6 Results and Analysis

6.1 RQ1: Language is Causally Load-Bearing and Scale-Invariant

Table 2 presents Task Success Rate (TSR) and Causal Sensitivity Score (CSS) across all evaluated model architectures under canonical (original), destructive (blank, nonsense), misleading (wrong_action, wrong_object, wrong_task), and control (repeated) prompt conditions (column definitions detailed in Table 2 caption).

Destructive conditions (blank, nonsense) collapse task success across all architectures. SmolVLA-450M drops to 0.0% success (CSS = 1.00). OpenVLA-7B drops to 0.0% under blank (CSS = 1.00). OpenVLA-OFT retains a minor residue (12.5% under blank, CSS = 0.87), which we attribute to visual shortcuts enabled by its wrist camera and proprioception inputs. Crucially, causal sensitivity is flat across model scale: CSS remains near 1.00 from 0.45B to 7.5B parameters.

6.2 RQ2: Information Structure Confounds Word-Class Grounding

Prior studies observe that object swaps collapse performance while verb swaps are tolerated, concluding that VLAs bind nouns strictly but ignore verbs. We find that this conclusion is confounded by the benchmark’s information structure.

We compute the conditional task entropy $H(\text{task} | W)$ over LIBERO-Goal’s ten task instructions under a uniform task prior ($H(\text{task}) = \log_2 10 = 3.32$ bits):

$$\begin{aligned} H(\text{task} | \text{noun set}) &= 0.00 \text{ bits}, \\ H(\text{task} | \text{verb set}) &= 1.55 \text{ bits}. \end{aligned} \quad (2)$$

All ten noun sets are unique across tasks, resolving all 3.32 bits of task identity. In contrast, the verb *put* appears in six of the ten tasks, leaving 1.55 bits unresolved (Figure 2). Conditional on knowing the nouns, the verb contributes *exactly zero bits* of task information.

This information redundancy has two major implications: 1. Ignoring the verb is informationally optimal on LIBERO-Goal. 2. Standard verb substitution probes are uninformative by construction:

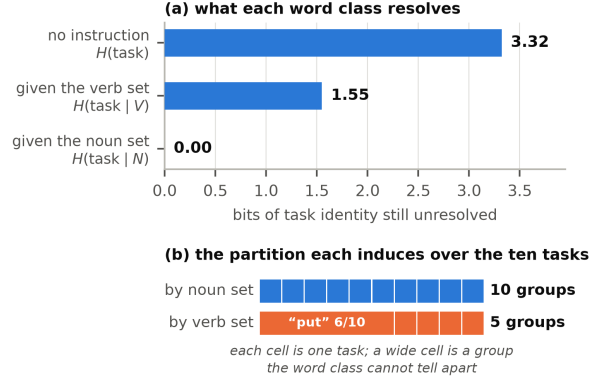


Figure 2: **Information Content of LIBERO-Goal Instructions.** (a) Conditional task entropy $H(\text{task} | W)$ under a uniform prior (3.32 bits total). Object nouns uniquely identify all ten tasks ($H = 0.00$ bits), rendering verbs redundant. (b) Verb frequency breakdown: the verb *put* dominates 6 out of 10 task instructions, creating an uninformative verb cluster.

7 of the 10 verb swaps replace *put* with *push*, leaving the noun set and task identity unchanged.

6.3 The Word-Class Removal Test

To test this entropy account, we perform a removal test: rather than substituting words, we drop word classes outright. Deleting the verb (0 bits destroyed) should leave performance unchanged, whereas masking nouns (3.32 bits destroyed) should collapse performance.

The empirical results confirm our prediction (Table 3, Figure 3). Deleting the verb changes success by only 2.0 points on OpenVLA-OFT (McNemar $p = 0.34$, 10 changed outcomes out of 200 matched pairs, consistent with random chance). Masking nouns costs 73.5 points and flips 148 pairs to failure. Two edits of similar edit distance differ by two orders of magnitude in impact because of their information content.

6.4 Re-evaluating Word-Class Asymmetry: Antonym vs. Near-Synonym Split

We separate the ten verb substitutions into the 3 genuine antonym pairs (*open* ↔ *close* twice, *turn on* ↔ *turn off* once) and the 7 near-synonym control pairs (Table 4, Figure 4).

Splitting the probe reverses the conclusion for OpenVLA-7B. Evaluating the suite-wide *wrong_action* probe yields a mild 12.0-point drop. However, splitting the probe reveals that OpenVLA-7B drops 31.7 points on true antonyms ($p < 10^{-5}$), while dropping only 3.6 points on

Model	Baseline	Blank	Nonsense	Wrong Action	Wrong Object	Wrong Task	Repeated
SmolVLA (0.45B)	78.5%±2.5	0.0% (1.00)	0.0% (1.00)	64.5%±0.5	9.3%±2.1	0.0%	10.0%±2.0
OpenVLA (7B)	71.0%±1.0	0.0% (1.00)	0.0% (1.00)	59.0%	10.7%±0.7	0.0%	58.5%±0.5
OpenVLA-OFT (7.5B)	97.5%±1.5	12.5%±2.5 (0.87)	14.0%±1.0 (0.86)	94.5%±0.5	11.4%	0.0%	96.5%±1.5
<i>episodes per cell</i>	200	200	200	200	140	200	200

Table 2: **Task Success Rate (TSR) and Causal Sensitivity Score (CSS) across Model Scales.** Comprehensive evaluation of SmolVLA-450M, OpenVLA-7B, and OpenVLA-OFT-7.5B under structural and semantic instruction perturbations on LIBERO-Goal. Columns evaluate: **Baseline** (I_{orig}), **Blank** (empty prompt), **Nonsense** (out-of-domain tokens), **Wrong Action** (verb swap), **Wrong Object** (noun swap), **Wrong Task** (full instruction swap to an alternate task in scene S_0), and **Repeated** (length control). Causal Sensitivity Scores ($\text{CSS} \in [0, 1]$) in parentheses demonstrate that language is causally load-bearing and scale-invariant across a $16\times$ parameter range. Episode counts per condition and model range from $n = 140$ to $n = 200$ per cell across evaluated scenes. All models pass the scene-fixed check on every comparable episode.

Condition	bits	TSR	Δ [95% CI]	p
<i>OpenVLA (7B)</i>				
original	0.00	71.0%	0.0	—
verb_dropped	0.00	75.0%	+4.0 [−3.5, +11.0]	0.36
nouns_masked	3.32	6.0%	−65.0 [−71.5, −58.0]	$< 10^{-10}$
blank	3.32	0.0%	−71.0 [−77.0, −64.5]	$< 10^{-10}$
<i>OpenVLA-OFT (7.5B)</i>				
original	0.00	97.5%	0.0	—
verb_dropped	0.00	95.5%	−2.0 [−5.0, +1.0]	0.34
nouns_masked	3.32	24.0%	−73.5 [−79.5, −67.5]	$< 10^{-10}$
blank	3.32	12.5%	−85.0 [−89.5, −80.0]	$< 10^{-10}$

Table 3: **Word-Class Removal Test.** Comparison of Task Success Rate (TSR) and drop (ΔTSR) when deleting action verbs (0 bits destroyed) versus masking object nouns (3.32 bits destroyed) on OpenVLA-7B and OpenVLA-OFT-7.5B ($n = 200$ per cell, p -values computed via McNemar’s test over matched initial scenes S_0). Coverage: OpenVLA (7B) on 10 of 10 tasks; OpenVLA-OFT (7.5B) on 10 of 10 tasks. The ablation conditions are generated in the same process as original, so the Δ column is a within-scene contrast.

near-synonyms ($p = 0.52$). OpenVLA-7B actively reads and grounds action verbs when the prompt contains true task contrast; suite-wide probes mask this by averaging real effects with uninformative controls.

6.5 Paraphrase Robustness and Kinematics

Table 5 reports PRIDE paraphrase scores. Robustness scales with parameter count: SmolVLA scores PRIDE 2.7, OpenVLA-7B scores 33.3, and OpenVLA-OFT achieves 65.8. All models follow the same vulnerability ordering: object paraphrases degrade performance least, while compositional paraphrases degrade performance most.

Kinematic analysis (t_{div} , threshold 5 cm) confirms these findings: under `blank` instructions, trajectories depart from canonical paths within 1–2 steps. Under `wrong_action`, trajec-

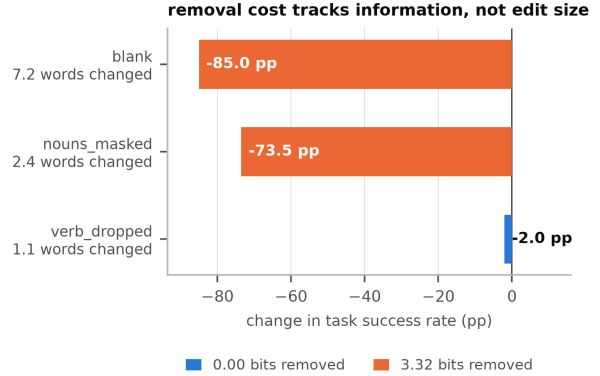


Figure 3: **Word-Class Removal Impact Tracks Information Content, Not Edit Size (OpenVLA-OFT-7.5B).** Change in Task Success Rate (ΔTSR in percentage points relative to baseline I_{orig}) when removing different word classes from prompt instructions. Colors indicate information content destroyed: **Blue** = 0.00 bits removed; **Orange** = 3.32 bits (full task entropy) removed. Deleting action verbs (`verb_dropped`, 1.1 words changed) removes 0 bits of task entropy and causes a statistically insignificant drop of only -2.0 pp ($p = 0.34$, McNemar test). In contrast, masking target nouns (`nouns_masked`, 2.4 words changed) destroys all 3.32 bits of task entropy and collapses performance by -73.5 pp, closely tracking full instruction removal (`blank`, 7.2 words changed, -85.0 pp). This demonstrates that VLA prompt sensitivity is governed by information theory rather than raw edit distance.

ries follow canonical paths for 6–11 steps before diverging, matching the timing expected when nouns specify target approach location and verbs specify manipulation manner. Qualitative rollout snapshots across key timesteps ($t = 0, 75, 150, 225, 300$) are visualized in Figure 5.

7 Discussion and Conclusion

We evaluated three VLA architectures under a scene-fixed causal protocol to determine the na-

Model	antonym		near-synonym	
	Δ TSR	p	Δ TSR	p
SmolVLA (0.45B)	-21.7 pp	10^{-3}	-10.7 pp	0.02
OpenVLA (7B)	-31.7 pp	10^{-6}	-3.6 pp	0.52
OpenVLA-OFT (7.5B)	-5.0 pp	0.45	-2.1 pp	0.51

Table 4: **Verb Substitution Split: Antonyms vs. Near-Synonyms.** Task Success Rate drop (Δ TSR) and McNemar statistical significance (p -values) when splitting verb substitutions into contrastive antonym pairs ($n = 60$) versus near-synonym control pairs ($n = 140$). OpenVLA-7B exhibits significant verb sensitivity (31.7 pp drop, $p < 10^{-5}$) only when verbs define distinct task goals.

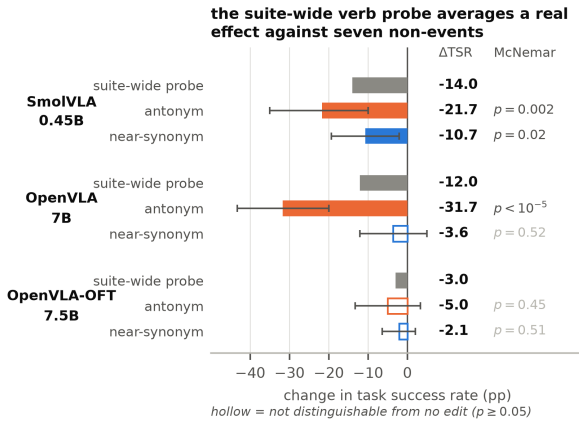


Figure 4: **Task Success Rate Drop Under Contrastive Antonym vs. Near-Synonym Verb Substitutions.** Performance change (Δ TSR) on matched pairs for SmolVLA, OpenVLA-7B, and OpenVLA-OFT-7.5B. OpenVLA-7B exhibits a highly significant 31.7 pp drop ($p < 10^{-5}$) on genuine antonym pairs (open $\leftrightarrow$ close, turn on $\leftrightarrow$ turn off), proving that it reads verbs when the benchmark requires it.

ture of language grounding failures. Our findings establish three key conclusions:

1. **Language is causally necessary:** Completely removing language collapses task success across all scales, disproving the notion that VLAs operate as purely visual policies on shared-scene benchmarks.
2. **Benchmark information redundancy creates apparent verb blindness:** In LIBERO-Goal, nouns uniquely determine task identity. Models ignore verbs because verbs are redundant, not because models are incapable of verb grounding.
3. **VLAs ground verbs when prompts are contrastive:** On true verb antonyms, OpenVLA-7B exhibits strong verb sensitivity (31.7-point drop, $p < 10^{-5}$).

We recommend that future VLA benchmarks measure and report instruction set information content before drawing word-class grounding con-

clusions.

Limitations

Our study has several scope bounds: (1) **Single benchmark suite:** Causal isolation relies on LIBERO-Goal’s shared initial scene. (2) **Antonym task coverage:** Only three tasks in LIBERO-Goal afford true verb antonyms ($n = 60$ episodes across 3 task templates). (3) **Simulation scope:** All experiments take place in MuJoCo simulation without real-robot deployment. (4) **Uniform entropy prior:** Information calculations assume a uniform task distribution, which may differ from model pre-training data distributions.

Ethics and Reproducibility

This work uses simulated environments and open-source model checkpoints. We release per-run manifests (environment locks, state hashes) to ensure full computational reproducibility.

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Paraphrase Axis	n	Scenes	TSR	Δ TSR (pp)	PRIDE Score
<i>SmolVLA (0.45B)</i>					
Object	259	20	29.73%	-48.77	28.7
Action	870	20	7.24%	-71.26	4.8
Compositional	2963	20	1.79%	-76.71	1.5
All axes	4092	24	4.72%	-73.78	2.7
<i>OpenVLA (7B)</i>					
Object	150	150	48.00%	-23.00	36.3
Action	150	150	60.67%	-10.33	51.8
Compositional	150	150	24.00%	-47.00	20.2
All axes	450	150	44.22%	-26.78	33.3
<i>OpenVLA-OFT (7.5B)</i>					
Object	150	150	78.00%	-19.50	72.5
Action	150	150	90.67%	-6.83	86.3
Compositional	150	150	54.00%	-43.50	50.3
All axes	450	150	74.22%	-23.28	65.8

Table 5: **Paraphrase Robustness and PRIDE Metric Suite.** Comprehensive evaluation of paraphrase sensitivity across Object, Action, and Compositional rewording axes ($n = 4092$ for SmolVLA, $n = 450$ for OpenVLA/OpenVLA-OFT). PRIDE scores combine task success with keyword and structural similarity distance ($PD = 1 - 0.5(SK + ST)$), demonstrating that robustness scales monotonically with model capacity.

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Qualitative Rollout Snapshots Across Instruction Conditions

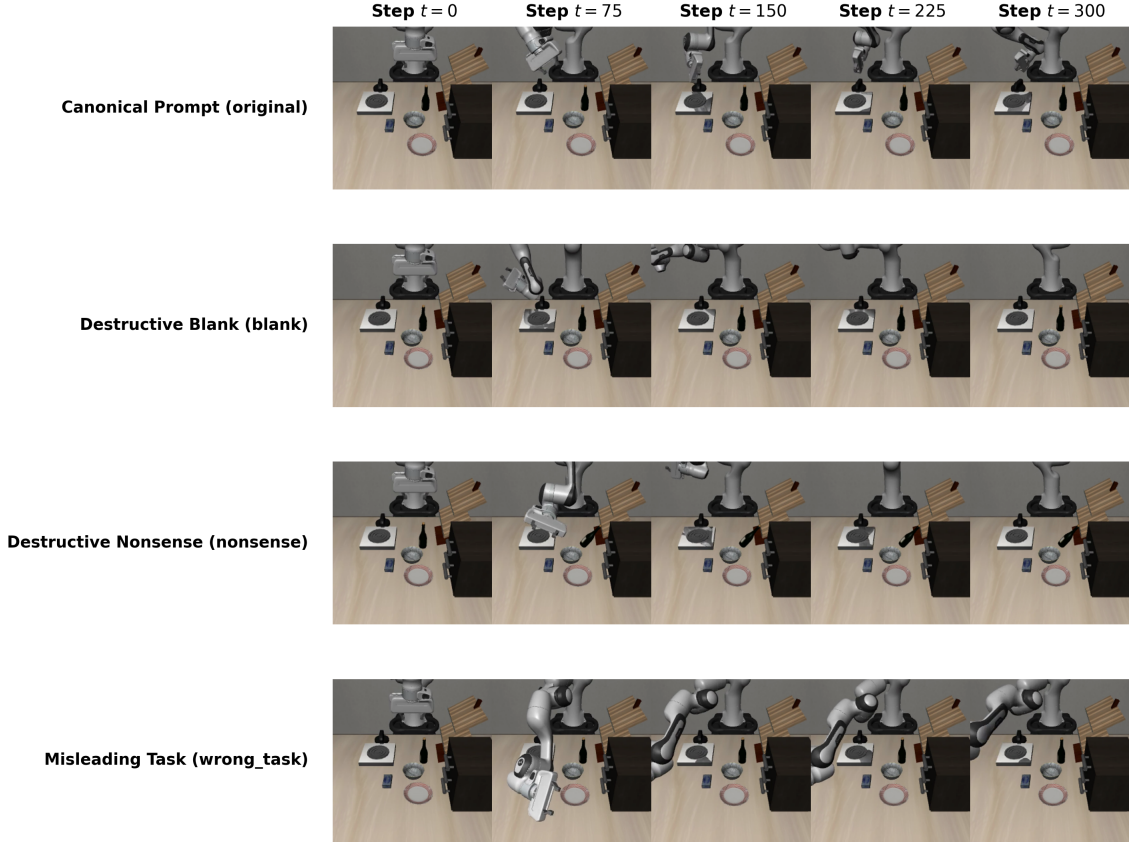


Figure 5: **Qualitative Rollout Snapshot Comparison Across Timesteps.** Temporal frame snapshots ($t = 0, 75, 150, 225, 300$) of MuJoCo simulator rollouts under canonical, destructive (blank, nonsense), and misleading (`wrong_task`) instructions on LIBERO-Goal. Highlights physical divergence behavior, in-place hesitation, and task switching.

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A Full Per-Cell Results

Table 6 provides per-cell episode counts and 95% Wilson confidence intervals for all RQ1 conditions. Table 7 reports kinematic divergence metrics.

B Scene-Fixed Invariant Audit

Standard LIBERO reset logic advances an internal generator across calls. As a result, sequential conditions run in separate processes silently receive different initial object poses. We enforce per-task

process execution and validate state hashes (`qpos`, `qvel`) to guarantee strict scene matching across conditions.

C PRIDE Metric Formulation

PRIDE (Kim et al., 2026) weights paraphrase success by distance from canonical instructions: $PD_i = 1 - 0.5(SK_i + ST_i)$ and $PRIDE = 100 \sum_i s_i PD_i / \sum_i PD_i$.

D Kinematic Divergence Analysis

Figure 6 illustrates end-effector divergence $e(t)$ over time step t .

Model	Condition	n	TSR (95% CI)	CSS
SmolVLA (0.45B)	original	200	78.5% [72.3, 83.6]	–
SmolVLA (0.45B)	blank	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	nonsense	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	wrong_action	200	64.5% [57.7, 70.8]	–
SmolVLA (0.45B)	wrong_object	140	9.3% [5.5, 15.2]	–
SmolVLA (0.45B)	wrong_task	200	0.0% [0.0, 1.9]	–
SmolVLA (0.45B)	repeated	200	10.0% [6.6, 14.9]	–
OpenVLA (7B)	original	200	71.0% [64.4, 76.8]	–
OpenVLA (7B)	blank	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	nonsense	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	wrong_action	200	59.0% [52.1, 65.6]	–
OpenVLA (7B)	wrong_object	140	10.7% [6.6, 16.9]	–
OpenVLA (7B)	wrong_task	200	0.0% [0.0, 1.9]	–
OpenVLA (7B)	repeated	200	58.5% [51.6, 65.1]	–
OpenVLA-OFT (7.5B)	original	200	97.5% [94.3, 98.9]	–
OpenVLA-OFT (7.5B)	blank	200	12.5% [8.6, 17.8]	0.87
OpenVLA-OFT (7.5B)	nonsense	200	14.0% [9.9, 19.5]	0.86
OpenVLA-OFT (7.5B)	wrong_action	200	94.5% [90.4, 96.9]	–
OpenVLA-OFT (7.5B)	wrong_object	140	11.4% [7.2, 17.8]	–
OpenVLA-OFT (7.5B)	wrong_task	200	0.0% [0.0, 1.9]	–
OpenVLA-OFT (7.5B)	repeated	200	96.5% [93.0, 98.3]	–

Table 6: **Complete Per-Cell RQ1 Experimental Results.** Detailed breakdown of Task Success Rates (TSR), 95% Wilson score confidence intervals, sample sizes n , and Causal Sensitivity Scores (CSS) across all 3 model architectures and 7 instruction conditions on LIBERO-Goal.

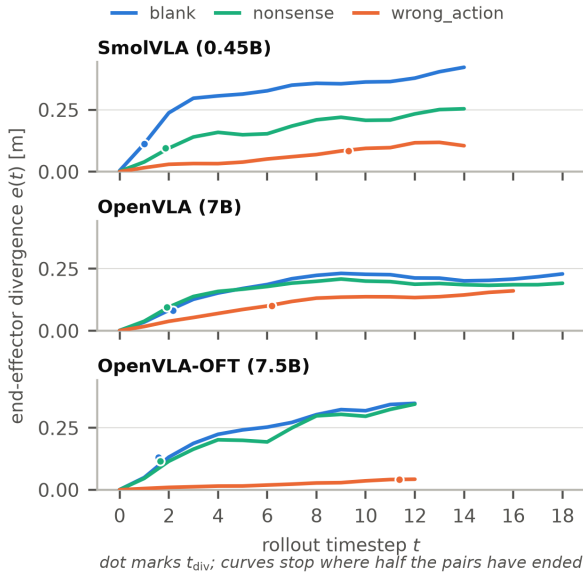


Figure 6: **End-Effector Trajectory Divergence Profiles $e(t)$ Over Rollout Timesteps.** Mean Euclidean distance (meters) between canonical and perturbed end-effector trajectories sharing identical initial state S_0 . Destructive prompts (blank, nonsense) diverge immediately at $t_{\text{div}} \leq 2$ steps, while wrong_action prompts track canonical trajectories for 6–12 steps during initial target approach before diverging during grasping.

Condition	Pairs	t_{div} (steps)	e_{10} [m]
<i>SmolVLA (0.45B)</i>			
blank	84	1.00	0.3548
nonsense	84	1.87	0.2197
para_action	10	2.00	0.1972
para_compositional	13	1.85	0.2086
para_object	13	4.69	0.1383
repeated	84	2.40	0.1737
wrong_action	77	9.31	0.0838
wrong_task	84	1.64	0.3278
wrong_object	56	1.96	0.1785
<i>OpenVLA (7B)</i>			
blank	73	2.16	0.2315
nonsense	73	1.92	0.2087
nouns_masked	73	3.11	0.2015
repeated	65	6.35	0.1082
verb_dropped	68	7.87	0.0764
wrong_action	68	6.18	0.1355
wrong_task	73	2.25	0.2992
wrong_object	48	2.75	0.1920
<i>OpenVLA-OFT (7.5B)</i>			
blank	55	1.58	0.3236
nonsense	54	1.65	0.3043
nouns_masked	55	2.67	0.2751
para_action	50	11.70	0.0339
para_compositional	55	6.58	0.1474
para_object	51	10.90	0.0582
repeated	43	12.02	0.0234
verb_dropped	53	12.13	0.0206
wrong_action	51	11.37	0.0289
wrong_task	55	1.64	0.3127
wrong_object	35	1.89	0.3173

Table 7: **Kinematic Trajectory Divergence Metrics.** Quantitative trajectory departure steps (t_{div} , threshold $\tau = 5$ cm) and 10-step end-effector spatial displacement (e_{10} in meters) between canonical and perturbed roll-outs sharing identical initial physical simulator states S_0 .